

KHAMBAMETTU MADHU SUDHAN RAO

+91 7981002778 | madhusudhan2k6@gmail.com | Portfolio | [linkedin.com/in/madhuu-k](https://www.linkedin.com/in/madhuu-k) | github.com/Madhuu-k

SUMMARY

CS Undergrad (CGPA: 9.4/10) specializing in high-efficiency Machine Learning and Full-Stack systems. Engineered CV pipelines achieving 30+ FPS on CPU-only hardware and architected Dockerized, SQL-driven applications. Focused on building production-grade software with optimized performance and resource constraints.

TECHNICAL SKILLS

Languages: Python (primary), SQL, JavaScript, C++, Java
ML/AI: TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib
Computer Vision: OpenCV, MediaPipe, Pose Estimation
Web/Infra: React, Flask, REST APIs, Docker, PostgreSQL, Git
Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks
Tools: Jupyter, Google Colab, Linux, VS Code

PROJECT EXPERIENCE

AI Fitness Trainer | *Python, OpenCV, MediaPipe* | GitHub Feb 2026

- Built a CPU-only pose estimation pipeline hitting 30+ FPS—eliminating GPU dependency and making real-time deployment viable on standard hardware.
- Designed an FSM-based repetition counter achieving 96% accuracy across 1,000+ frames.
- Reduced jitter and false detections using temporal smoothing techniques, ensuring stable real-world performance.
- Optimized inference to maintain <200 MB memory footprint—ensuring compatibility with low-resource edge devices.

Plant Disease Detection (Faculty Collaboration) | *Python, TensorFlow* | GitHub Jan 2026

- Developed a transfer-learning CNN under faculty guidance, achieving 94% accuracy on 5,000+ images.
- Mitigated dataset bias using stratified splitting and class balancing to improve generalization.
- Built feature embedding pipeline for similarity-based error analysis and interpretability.

Apex Stock | *React, Flask, PostgreSQL, Docker* | GitHub Nov 2025

- Architected a full-stack system with RBAC and normalized SQL—ensuring 100% data integrity and secure multi-user access.
- Automated procurement workflows using threshold-based restock logic and real-time reporting.
- Containerized application using Docker to ensure consistent deployment across environments.

EDUCATION

KL University <i>Bachelor of Technology in Computer Science, CGPA: 9.4/10</i>	Guntur, India 2023 – 2027 (<i>Expected</i>)
Sri Chaitanya Junior College <i>Intermediate (MPC), GPA: 9.3/10</i>	Vijayawada, India 2021 – 2023

CODING PROFILES

- **LeetCode:** leetcode.com/u/klu2300031210 | **200+ Problems Solved** (Focus: Arrays, Linked Lists, and Dynamic Programming)
- **GeeksforGeeks:** [geeksforgeeks.org/profile/madhusudtxfu](https://www.geeksforgeeks.org/profile/madhusudtxfu) | Consistently active in solving platform-specific challenges and exploring core DSA concepts.